CDA 106
Numerical Methods for Data Science

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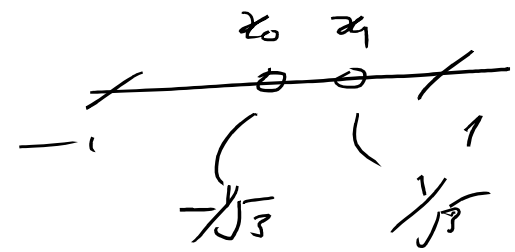
- Newton Cotes Methods (Equidistant points 'n+1' in number)

$$\int_a^b f(x)\,dx = \sum_{i=0}^{n} \lambda_i f(x_i)$$

Gauss Legendre Two points method



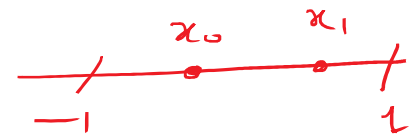
$$\int_{-1}^{1} f(x)\,dx = \lambda_0 f(x_0) + \lambda_1 f(x_1)$$

$\lambda_0 = 1$, $x_0 = -1/\sqrt{3}$, $\lambda_1 = 1$, $x_1 = 1/\sqrt{3}$

How did we get these specific points?

Derivation of Gauss Legendre 2 point Method

Consider $\int_{-1}^{1} f(x)\,dx = \lambda_0 f(x_0) + \lambda_1 f(x_1)$ — (1)



Here, $\lambda_0, \lambda_1, x_0, x_1$ are unknown.

Notice that $\lambda_0 \neq 0,\ \lambda_1 \neq 0.\ x_0 \neq x_1.\ -1 < x_0 < x_1 < 1$

Why only 4 functions

Important Assumption Assum (1) give exact value for

$f(x) = 1$, $f(x) = x$, $f(x) = x^2$, $f(x) = x^3$.

When $f(x) = 1$, (1) $\Rightarrow \int_{-1}^{1} 1\,dx = \lambda_0 \cdot 1 + \lambda_1 \cdot 1 \Rightarrow 2 = \lambda_0 + \lambda_1$

When $f(x) = x$ (1) $\Rightarrow \int_{-1}^{1} x\,dx = \lambda_0 x_0 + \lambda_1 x_1 \Rightarrow 0 = \lambda_0 x_0 + \lambda_1 x_1$

When $f(x) = x^2$, (1) $\Rightarrow \int_{-1}^{1} x^2\,dx = \lambda_0 x_0^2 + \lambda_1 x_1^2 \Rightarrow \frac{2}{3} = \lambda_0 x_0^2 + \lambda_1 x_1^2$

When $f(x) = x^3$ (1) $\Rightarrow \int_{-1}^{1} x^3\,dx = \lambda_0 x_0^3 + \lambda_1 x_1^3 \Rightarrow 0 = \lambda_0 x_0^3 + \lambda_1 x_1^3$

4 Eqns in 4 unknowns

$$\lambda_0 + \lambda_1 = 2 \quad \text{— (a)}$$

$$\lambda_0 x_0 + \lambda_1 x_1 = 0 \quad \text{— (b)} \checkmark$$

$$\lambda_0 x_0^2 + \lambda_1 x_1^2 = \frac{2}{3} \quad \text{— (c)}$$

$$\lambda_0 x_0^3 + \lambda_1 x_1^3 = 0 \quad \text{— (d)}$$

<u>Aim</u> To find $\lambda_0, \lambda_1, x_0, x_1$ i.e. to solve (a) – (d)

These are non-linear eqn!

→ Eliminate λ_0 from (b) & (d)

Multiply eqn (b) by x_0^2 and substract (d) from that

$$(b) \times x_0^2 - (d): \quad \lambda_0 \cancel{x_0^3} + \lambda_1 x_1 x_0^2 - \lambda_0 \cancel{x_0^3} - \lambda_1 x_1^3 = 0$$

$$\Rightarrow \lambda_1 x_1 (x_0^2 - x_1^2) = 0$$

- $\lambda_1 = 0$ — This is not possible
- $x_1 = 0$ — $(b) \Rightarrow \lambda_0 x_0 = 0$
  - $\lambda_0 = 0$ — This is not possible
  - $x_0 = 0$ — $x_1 = x_0 = 0$ NOT possible
- $x_0 = x_1$ — This is not possible
- $(x_0 = -x_1)$ — This is only possible

from the above we find $\boxed{x_0 = -x_1}$ or $\boxed{x_1 = -x_0}$.

$$(b) \Rightarrow \lambda_0 x_0 + \lambda_1(-x_0) = 0 \Rightarrow x_0(\lambda_1 - \lambda_0) = 0$$

$$\Rightarrow \boxed{\lambda_1 = \lambda_0} \quad \text{use in (a)}$$

$$\lambda_1 + \lambda_0 = 2 \Rightarrow \boxed{\lambda_0 = \lambda_1 = 1}$$

$$(c) \Rightarrow \lambda_0 x_0^2 + \lambda_1 x_1^2 = \frac{2}{3} \Rightarrow 2x_0^2 = \frac{2}{3} \Rightarrow x_0^2 = \frac{1}{3}$$

$$\Rightarrow x_0 = \pm \frac{1}{\sqrt{3}} \quad \Rightarrow x_1 = \mp \frac{1}{\sqrt{3}}$$

$x_0 = -\frac{1}{\sqrt{3}}$ $\quad x_1 = \frac{1}{\sqrt{3}}$ Correct choice

$x_0 = \frac{1}{\sqrt{3}}$ $\quad x_1 = -\frac{1}{\sqrt{3}}$ ✗

finally, $\lambda_0 = 1$, $\lambda_1 = 1$

$x_0 = -\frac{1}{\sqrt{3}}$ $\quad x_1 = \frac{1}{\sqrt{3}}$

$$\boxed{\int_{-1}^{1} f(x)\,dx = f\left(-1/\sqrt{3}\right) + f\left(1/\sqrt{3}\right)}$$

This is our Gauss Legendre two point method

In the similar fashion we can develop Gauss Legendre three-point.

$$\int_{-1}^{1} f(x)\,dx = \lambda_0 f(x_0) + \lambda_1 f(x_1) + \lambda_2 f(x_2)$$

$$\lambda_0 = \frac{5}{9}, \quad \lambda_2 = \frac{5}{9}, \quad \lambda_1 = 8/9$$

$$x_0 = -\sqrt{\frac{3}{5}} \qquad x_1 = 0 \qquad x_2 = \sqrt{\frac{3}{5}}.$$

Now, I shall Introduce you to <u>specific function</u>

called <u>Legendre Polynomial</u>. $(P_n(x))$ n degree

$\underline{n \geq 0}$

$$-P_n(x) = \frac{1}{2^n \cdot n!} \frac{d^n}{dx^n}(x^2-1)^n$$ Rodrigue's formula

$\underline{n=0}$ $\quad P_0(x) = 1$

$n=1$ $\quad P_1(x) = \frac{1}{2 \cdot 1} \frac{d}{dx}(x^2-1) = x$

$n=2$ $\quad P_2(x) = \frac{1}{2^2 \cdot 2!} \frac{d^2}{dx^2}(x^2-1)^2 = \frac{1}{8}\left[\frac{d}{dx}\left(2(x^2-1)\,2x\right)\right] = \frac{1}{8}\left[12x^2-4\right]$

$P_2(x) = \frac{3x^2-1}{2}$

$\underline{n=3}$ $\quad P_3(x) = \frac{1}{2}(5x^3-3x)$

$P_n(x)$ are sol. of Legendre diff eqn

$$(1-x^2)y'' - 2xy' + n(n+1)y = 0$$

Now, look at the zeros (roots) of

$$P_2(x) = 0 \Rightarrow \frac{3x^2-1}{2} = 0 \Rightarrow x = \pm \frac{1}{\sqrt{3}}$$

Roots of

$$P_3(x) = 0 \Rightarrow \frac{5x^3-3x}{2} = 0 \Rightarrow x = 0 \ \& \ x = \pm\sqrt{3/5}$$

Therefore, there must be something about these polynomials.

$$\Downarrow$$

These polynomials $P_n(x)$ are orthogonal on $[-1,1]$

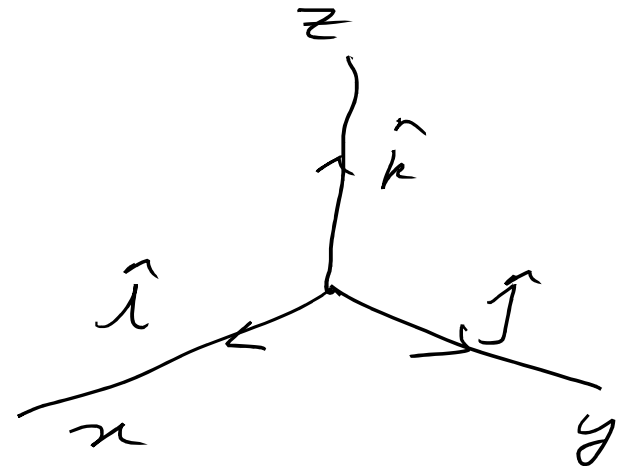
Que Are there other such polynomials which can be used for the approximation of Integrals ?? YES, "Hermite"

Orthogonality

$\hat{i}\cdot\hat{j}=0 \quad \hat{j}\cdot\hat{k}=0 \quad \hat{k}\cdot\hat{i}=0$

$\underline{a}\cdot\underline{b} = ab\cos\theta \quad \theta = 90^\circ$

$a\cdot b = 0 \Rightarrow$



When dot product is zero, vectors are $\perp$.

Orthogonal functions

"S" $= \{1, \sin x, \cos x, \sin 2x, \cos 2x, \sin 3x, \cos 3x, \ldots\}$

$$f\cdot g = \int_{-\pi}^{\pi} f(x)\,g(x)\,dx$$

If for any $f, g \in S$, $f \neq g$ $\quad f\cdot g = 0$

then we say f is orthogonal to g.

If it is true for every f & g then S is orthogonal set.

$\{P_n(x)\}_{n \geq 0}$ is also orthogonal set

$$\int_{-1}^{1} P_n(x)\, P_m(x)\, dx = 0 \qquad m \neq n$$

verify $\displaystyle\int_{-1}^{1} P_2(x) P_3(x)\, dx = 0$

LHS $\displaystyle\int_{-1}^{1} \frac{(3x^2-1)}{2} \times \frac{1}{2}(5x^3-3x)\, dx = 0$ RHS ✓

Hermite Polynomials

$$H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n}\left(e^{-x^2}\right) \qquad n \geq 0$$

$x \in (-\infty, \infty)$

$n=0 \qquad H_0 = 1$

$n=1 \qquad H_1 = 2x$

$n=2 \qquad H_2(x) = 4x^2 - 2$

$n=3 \qquad H_3(x) = 8x^3 - 12x$

Roots of $H_2(x) = \pm\frac{1}{\sqrt{2}}$

Roots of $H_3(x) = 0, \pm\sqrt{\frac{3}{2}}$

$\pm\frac{\sqrt{6}}{2}$

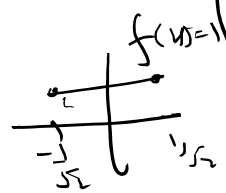
$$\int_{-\infty}^{\infty} e^{-x^2} H_n(x) H_m(x)\, dx = 0 \qquad m \neq n$$

Gauss Hermite 2 point formula

$$(*)\quad \boxed{\int_{-\infty}^{\infty} e^{-x^2} f(x)\,dx = \frac{\sqrt{\pi}}{2}\left[f\left(-\frac{1}{\sqrt{2}}\right) + f\left(\frac{1}{\sqrt{2}}\right)\right]}$$

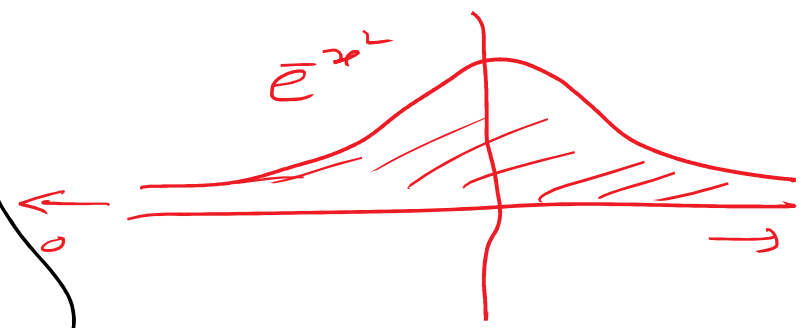
Example Find $\int_{-\infty}^{\infty} e^{-x^2} \cdot 1\,dx = ?$

Compare it $(*)$ $\quad f(x) = 1$



$$\int_{-\infty}^{\infty} e^{-x^2}\,dx = \frac{\sqrt{\pi}}{2}\left[1 + 1\right] = \sqrt{\pi}$$

Gaussian



Gauss Hermite 3 point

$$\int_{-\infty}^{\infty} e^{-x^2} f(x)\,dx = \frac{\sqrt{\pi}}{6}\left[f\left(-\sqrt{6}/2\right) + 4f(0) + f\left(\frac{\sqrt{6}}{2}\right)\right]$$

Ex $\int_{-\infty}^{\infty} e^{-x^2}\,dx = ?$ $\quad f(x) = 1$

$$\int_{-\infty}^{\infty} e^{-x^2} \cdot 1\,dx = \frac{\sqrt{\pi}}{6}\left[1 + 4 + 1\right] = \sqrt{\pi}$$

Ans.